

WASP-52b

R-0516

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_230806 · created 2026-07-20T22:27:34+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460162.8817 +/- 0.003 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.185 +/- 0.011
Transit depth (Rp/Rs)^2	3.41 +/- 0.39 [%]
Orbital Inclination (inc)	89.82 +/- 1.41
Airmass coefficient 1 (a1)	0.911 +/- 0.014
Airmass coefficient 2 (a2)	0.083 +/- 0.054
Scatter in the residuals of the lightcurve fit is	1.56 %
Best Comparison Star	#1 - [155, 65]
Optimal Method	PSF photometry
Transit Duration (day)	0.0888 +/- 0.0016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.185 ± 0.011 vs 0.1646 (deviation 1.67σ)
Duration fit vs published	0.0888 d vs 0.0758 d (deviation 7.28σ)
Mid-time O-C	-4.9 min
Depth SNR	15.1
Residual scatter	1.56 %

Light curve

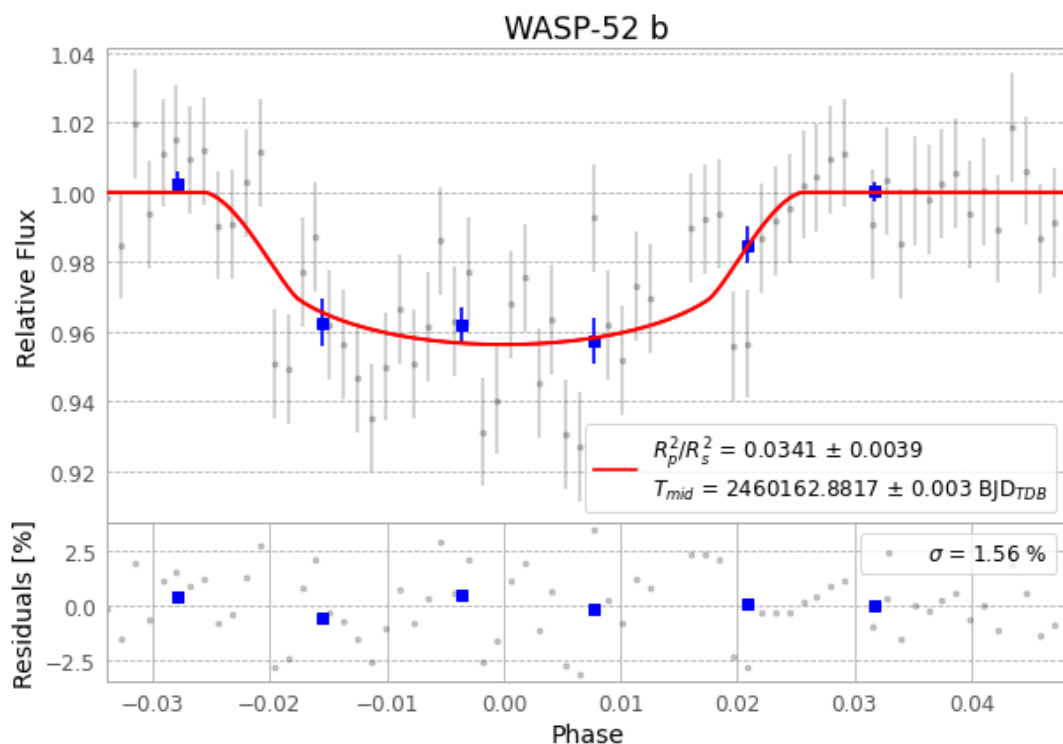


Figure 1 — FinalLightCurve_WASP-52 b_2023-08-06.png (SHA-256 88e762fe6658165e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [275, 186], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 8cdcd1250131bc99...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

69 FITS frames (45 MB), WASP-52230806073615.FITS → WASP-52230806110315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-230806.log (c2f6b09fd7b1...), FinalLightCurve_WASP-52 b_2023-08-06.png (88e762fe6658...), FinalLightCurve_WASP-52 b_2023-08-06.csv (a80ae7645656...)

Compute resources

Wall time 125.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-20 22:27Z.